



CANDIDATE – PLEASE NOTE!

PRINT your name on the line below and return this booklet with your answer sheet. Failure to do so may result in disqualification.

TEST CODE **01207010**

FORM TP 2014044

MAY/JUNE 2014

**CARIBBEAN EXAMINATIONS COUNCIL
CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION
BIOLOGY**

Paper 01 - General Proficiency

1 hour 15 minutes

11 JUNE 2014 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

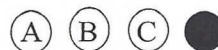
1. This test consists of 60 items. You will have 1 hour and 15 minutes to answer them.
2. In addition to this test booklet, you should have an answer sheet.
3. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.
4. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

Which of the following diseases is due to a dietary deficiency?

- (A) Malaria
- (B) Diabetes
- (C) Influenza
- (D) Anaemia

Sample Answer



The correct answer to this item is “Anaemia”, so answer space (D) has been shaded.

5. If you want to change your answer, erase it completely and fill in your new choice.
6. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to this item later. Your score will be the total number of correct answers.
7. Figures are not necessarily drawn to scale.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

- 2 -

1. A plant which has reduced leaves with no chlorophyll, lacks a proper root system, and flowers abundantly, is MOST likely to be found
- (A) in a habitat where water is scarce
 - (B) living parasitically on another plant
 - (C) growing in an area where there are few insects
 - (D) existing as a commensal with other green plants

Item 2 refers to the following food chain which shows the feeding relationships in a freshwater habitat.

Microscopic algae → mosquito larvae → small fish → large fish

2. The organisms to which the LEAST amount of energy is available is the
- (A) microscopic algae
 - (B) mosquito larvae
 - (C) small fish
 - (D) large fish

Item 3 refers to the following list.

- I. Number of legs
- II. Body temperature
- III. Shape of body
- IV. Type of body covering

3. The dead remains of a previously unknown animal are found in a forest. Which of the features listed above could be used to classify the animal?

- (A) I and II only
- (B) III and IV only
- (C) I, III and IV only
- (D) II, III and IV only

4. Which of the following types of bacteria is important for returning nitrogen gas to the atmosphere?

- (A) Nitrifying
- (B) Denitrifying
- (C) Putrefying
- (D) Nitrogen-fixing

5. A mature plant cell is different from a mature animal cell because the plant cell has

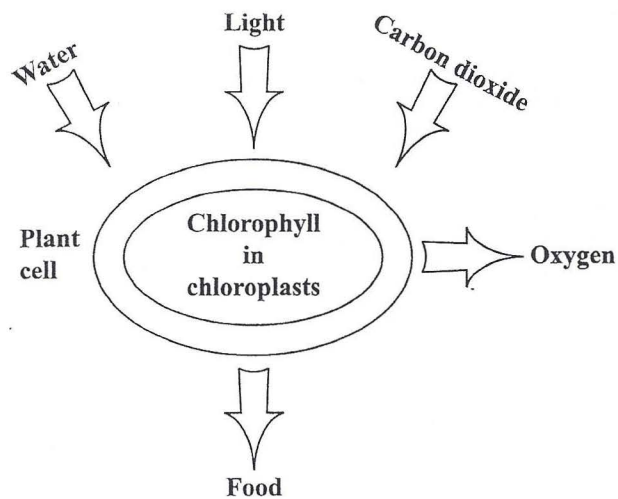
- (A) mitochondria
- (B) a cell membrane
- (C) a large permanent vacuole
- (D) a nucleus suspended in the cytoplasm

6. In a comparison between osmosis and diffusion, which of the following statements is NOT true?

- (A) Osmosis occurs only in living cells.
- (B) Both processes require energy to move particles across a concentration gradient.
- (C) Diffusion does not always require a partially permeable membrane.
- (D) With both processes, molecules move from a high concentration to an area of lower concentration.

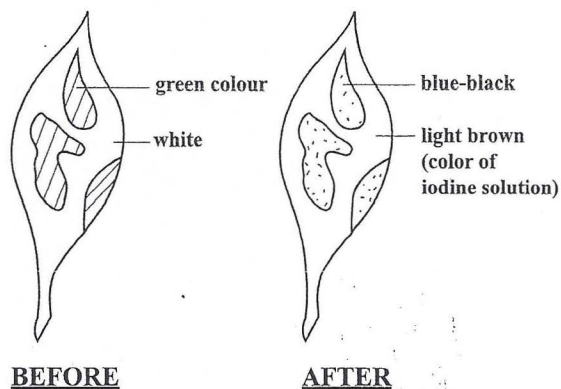
7. Which of the following organelles is the site of aerobic respiration?
- (A) Nucleus
 - (B) Cytoplasm
 - (C) Chloroplast
 - (D) Mitochondrion

Item 8 refers to the following diagram which represents a metabolic process carried out in plants.



8. The food produced is
- (A) fat
 - (B) starch
 - (C) protein
 - (D) amino acid
9. Which of the following is NOT an autotroph?
- (A) Fungus
 - (B) Moss
 - (C) Fern
 - (D) Alga

Item 10 refers to the following experiment. A green and white variegated leaf is exposed to sunlight for three hours and then tested for starch. The results of the test are shown in the diagram below.



10. Based on the experiment, which of the following conditions is necessary for photosynthesis?
- (A) Light
 - (B) Water
 - (C) Chlorophyll
 - (D) Carbon dioxide
11. When enzymes are boiled they are unable to function. This is because an increase in temperature
- (A) destroys food materials
 - (B) increases the rate of enzyme action
 - (C) denatures the protein of enzymes
 - (D) converts complex sugars to simple sugars
12. Which of the following occurs as a result of a deficiency of nitrogen in plants?
- (A) Yellowing of the older leaves of the plants
 - (B) Stunted chlorotic plants with reduced yields
 - (C) Greater susceptibility to diseases and insect attacks
 - (D) Dark green overly vigorous foliage leading to reduced yields

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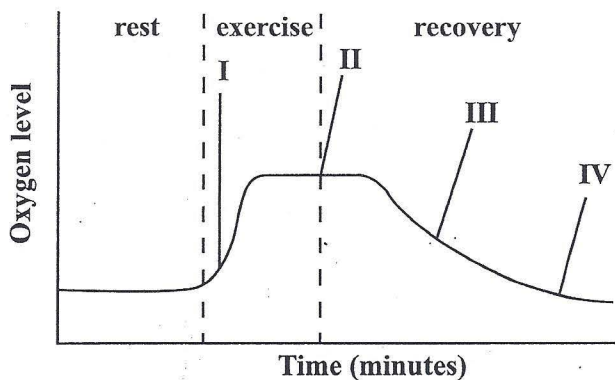
13. Which of the following is a function of hydrochloric acid in the stomach?

- (A) Emulsifying fat
- (B) Breaking down starch
- (C) Hydrolysing complex sugars
- (D) Providing a medium for pepsin

14. The role of respiration is BEST described as the

- (A) release of energy
- (B) absorption of oxygen
- (C) breakdown of carbohydrates
- (D) liberation of carbon dioxide

Item 15 refers to the following graph which shows the oxygen uptake of an athlete at rest, during exercise and after a period of exercise (recovery).



15. The HIGHEST concentration of lactic acid in the athlete's muscles occurs at

- (A) I
- (B) II
- (C) III
- (D) IV

16. Which of the following is NOT a reason for placing green plants in an aquarium?

- (A) Providing food for the animals
- (B) Providing oxygen for the animals
- (C) Removing the chlorine from the water
- (D) Removing the animals' waste products

17. Which of the following is an effect of nicotine found in cigarette smoke?

- (A) Increased heart rate
- (B) Increased beating of cilia
- (C) Reduced oxygen transport by the blood
- (D) Increased mucus production in the cells lining the respiratory passage

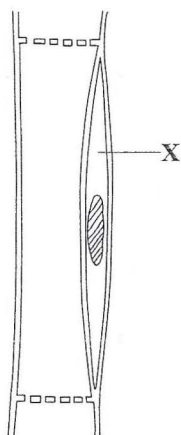
18. The right ventricle collects blood from the

- (A) aorta
- (B) vena cava
- (C) right atrium
- (D) pulmonary artery

19. An *Amoeba* obtains all the oxygen it needs by diffusion via its cell membrane, while a human needs to have special respiratory surfaces for this purpose. The BEST reason for this difference is that

- (A) the *Amoeba* does not require much oxygen
- (B) oxygen cannot pass through the skin of a human
- (C) a human requires a larger volume of oxygen
- (D) a human's surface area to volume ratio is too small for diffusion to be effective

Item 20 refers to the following diagram which shows the longitudinal section of the stem of a plant.



20. The part labelled X is the

- (A) cork cell
- (B) cambial cell
- (C) cortical cell
- (D) companion cell

21. Under which conditions will the rate of transpiration in plants be HIGHEST?

- (A) Sunny and low wind speed
- (B) Sunny and high wind speed
- (C) Cloudy and low wind speed
- (D) Cloudy and high wind speed

22. Mineral salts in plants are transported MAINLY through the

- (A) cambium
- (B) sieve tubes
- (C) parenchyma
- (D) xylem vessels

23. Storage organs in plants provide food

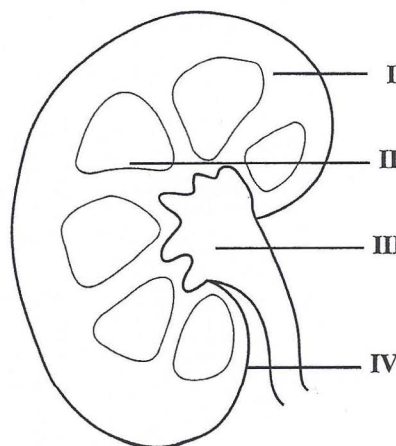
- I. for vegetative reproduction
- II. for the developing embryo
- III. to ensure survival when nutrients are scarce

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

24. Which of the following is NOT a form of excretion in plants?

- (A) Carbon dioxide diffusing in, through the stomata
- (B) Substances being stored in the bark of a tree
- (C) Water vapour diffusing out, through the epidermis
- (D) Stored calcium oxalate crystals in leaves during leaf fall

Item 25 refers to the following diagram of a mammalian kidney.



25. Into which structure is urine emptied?

- (A) I
- (B) II
- (C) III
- (D) IV

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26. Which of the following organs is MOST directly affected when the pituitary gland is malfunctioning?

(A) Kidney
(B) Lung
(C) Skin
(D) Colon

27. Which of the following is NOT a feature of the leaf of a xerophyte?

(A) Stomata abundant on upper surface
(B) Needle-like leaves
(C) Plugging of stomata
(D) Heavily cutinized

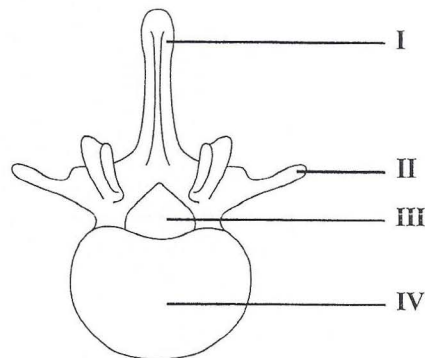
28. The MAIN function of cartilage at the end of long bones is to

(A) protect nerves
(B) reduce friction
(C) lubricate joints
(D) increase flexibility

29. Which of the following correctly describes movement in plants?

(A) Irreversible, whole or part movement
(B) Reversible, whole or part movement
(C) Irreversible, growth or part movement
(D) Reversible, growth or part movement

Item 30 refers to the following diagram which shows the structure of a typical vertebra.



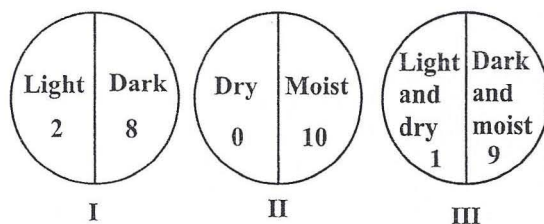
30. Which of the following shows the correct labelling of the parts?

	Neural canal	Centrum	Transverse process
(A)	I	II	III
(B)	II	III	IV
(C)	III	IV	II
(D)	IV	III	I

31. Which of the following is NOT true of locomotion?

(A) It allows organisms to find mates.
(B) It is in response to a stimulus.
(C) It allows organisms to find food.
(D) It is a characteristic of all living things.

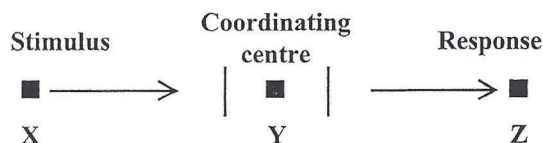
Item 32 refers to the following results of an investigation of how woodlice respond to light and moisture. The results show the number of woodlice that moved to varying conditions – light, dark, moist and dry.



32. The above responses are important to the woodlice in order to

- (A) ensure that they are close to a food supply
- (B) ensure that they do not become dehydrated
- (C) provide the best conditions for enzyme activity
- (D) provide protection from predators

Item 33 refers to the following diagram which shows a coordinated response to a stimulus in humans.



33. The role of Y in the illustration above is to

- (A) stimulate receptors
- (B) cause reflex actions
- (C) react to stimuli
- (D) interpret changes in the environment

34. For a student to recall his biology notes, action takes place in

- (A) a reflex arc
- (B) the cerebrum
- (C) the cerebellum
- (D) the medulla oblongata

35. Which hormone causes a person to show superhuman (enhanced) strength in an emergency situation?

- (A) Insulin
- (B) Adrenaline
- (C) Thyroxine
- (D) Pituitary hormone

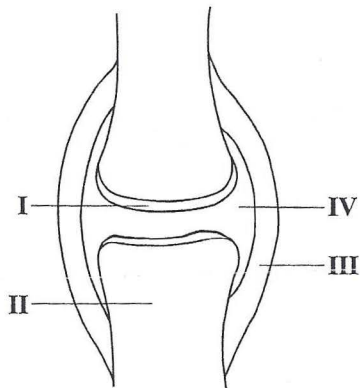
36. Which three structures of the skin are involved in temperature regulation?

- (A) Sweat gland, erector muscle, capillaries
- (B) Sebaceous gland, capillaries, temperature receptors
- (C) Erector muscle, temperature receptors, sweat gland
- (D) Temperature receptors, capillaries, erector muscle

37. The function of the choroid layer in the eye is to

- (A) focus most light rays
- (B) prevent internal reflection
- (C) maintain the shape of the eyeball
- (D) control the amount of light entering the eye

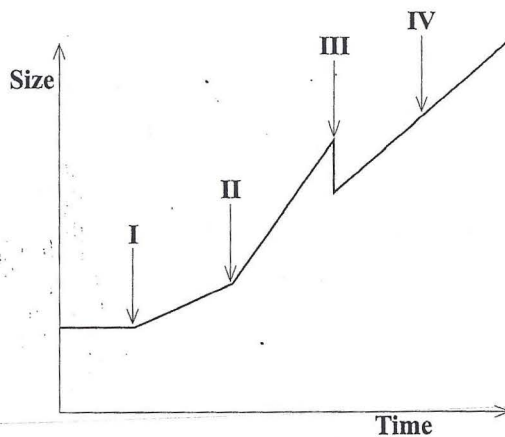
Item 38 refers to the following diagram of a section through a joint.



38. Which of the above labelled components reduce friction?

- (A) I and II
- (B) I and IV
- (C) II and III
- (D) II and IV

Item 39 refers to the following growth curve for an instar (young insect) during one stage of its life cycle.



39. At which point does moulting occur?

- (A) I
- (B) II
- (C) III
- (D) IV

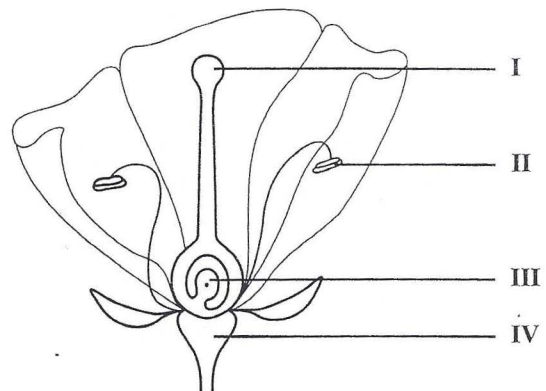
Item 40 refers to the following activities involved in the menstrual cycle.

- I. Repair of the uterus lining
- II. Ovulation
- III. Shedding of the uterus lining
- IV. Development of the Graafian follicle

40. Which of the following is the correct sequence of activities?

- (A) I, II, III, IV
- (B) II, I, III, IV
- (C) IV, I, II, III
- (D) IV, III, II, I

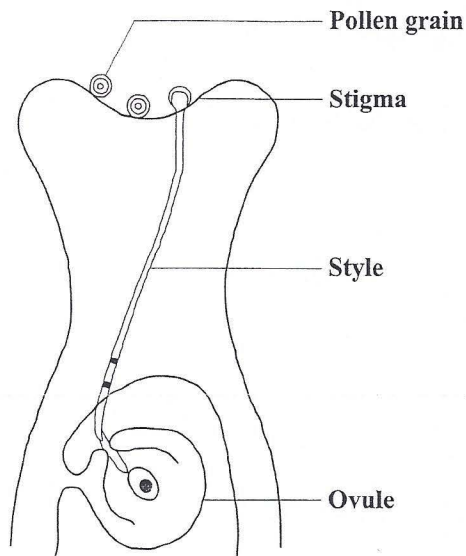
Item 41 refers to the following diagram of a flower.



41. Which of the labelled parts is responsible for the production of male gametes?

- (A) I
- (B) II
- (C) III
- (D) IV

Item 42 refers to the following diagram.



42. The MOST appropriate heading for this diagram is

- (A) ovulation
- (B) fertilization
- (C) pollination
- (D) reproduction

43. During pollination, pollen grains which are small, light, odourless and smooth are MOST likely carried by

- (A) wind
- (B) birds
- (C) water
- (D) insects

44. Contraception is used PRIMARILY to

- (A) limit the size of families
- (B) cause infertility in humans
- (C) prevent the spread of sexually transmitted infections
- (D) increase the world's population

45. Which of the following statements is true of wind dispersed seeds?

- (A) They are found in succulent fruit.
- (B) They have hooks which allow them to be taken great distances.
- (C) They are rapidly strewn from dry pods.
- (D) They have feathery hairs which allow them to be taken great distances.

46. Which of the following events does NOT occur during mitosis?

- (A) The chromatids exchange genetic information.
- (B) Chromosome makes an exact copy of itself.
- (C) Two chromatids are joined by a centromere.
- (D) Chromosomes align themselves on the equator of the spindle.

47. In which of the following does meiosis occur?

- (A) A developing embryo
- (B) The skin of a mammal
- (C) The testis of a mammal
- (D) The tip of the shoot of a plant

48. A girl has blood group A and her brother has blood group B. Her mother is blood group AB. They both have the same father. Which of the following are possible genotypes of the father?

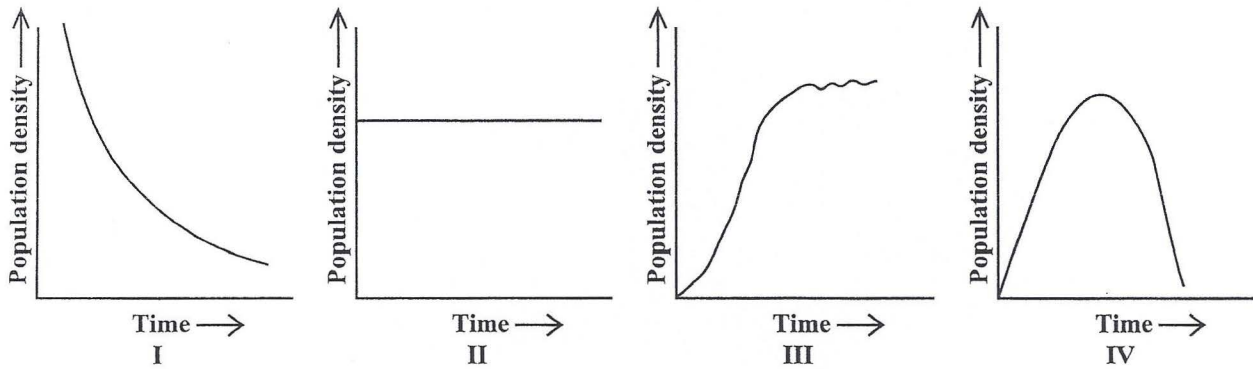
- I. AA
- II. BB
- III. AB
- IV. OO

- (A) I or II only
- (B) I or IV only
- (C) II or III only
- (D) III or IV only

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49. A plant breeder grows a new variety of tomato which has desirable characteristics. To develop this variety he sometimes encloses single flowers in bags. Which of the following is MOST likely a reason for this?
- (A) To ensure fertilization
 - (B) To ensure pollination
 - (C) To prevent damage by rain
 - (D) To prevent entry of pollen from other plants
50. Which of the following is an ecological implication of genetic engineering?
- (A) Loss of biodiversity
 - (B) Humans 'playing God'
 - (C) Higher prices for organic foods
 - (D) Reduced incidence of allergic reactions
51. In which of the following ways can hypertension be treated or controlled?
- I. Eating iron-rich foods
 - II. Low carbohydrate diet
 - III. Low salt intake
 - IV. Exercise
- (A) I and II only
 - (B) II and III only
 - (C) III and IV only
 - (D) I, II, III and IV
52. Abuse of prescription drugs is considered a disease because
- (A) it is costly to families
 - (B) it reduces human productivity
 - (C) many people are involved in it
 - (D) it affects the functioning of the body
53. Which of the following is a sign of AIDS?
- (A) Sterility
 - (B) Skin rash
 - (C) Blister on genitals
 - (D) Yellow discharge from penis
54. A vector is defined as an organism that
- (A) bites human
 - (B) causes diseases
 - (C) lives on other organisms
 - (D) transmits disease organisms
55. A boy stepped on a nail in the garden. He is rushed to the hospital where he is given antiserum. This provides
- (A) artificial passive immunity
 - (B) natural passive immunity
 - (C) artificial active immunity
 - (D) natural active immunity
56. A group of fish, snails, watergrass and tadpoles living in a freshwater pond may be described as a
- (A) niche
 - (B) habitat
 - (C) community
 - (D) population
57. The mark, release, capture technique would MOST likely be used to determine
- (A) lichens on a tree trunk
 - (B) watergrass on a football field
 - (C) the population of plankton in a pond
 - (D) the population of snails in a vegetable plot

Item 58 refers to the following graphs.



58. Which graph BEST represents changes in population density of a natural animal population, with respect to time?
- (A) I
(B) II
(C) III
(D) IV
59. Which of the following is NOT an example of a renewable energy resource?
- (A) Oil
(B) Wind
(C) Solar
(D) Biomass
60. Which of the following practices does NOT help to conserve the environment?
- (A) Overgrazing
(B) Crop rotation
(C) Reforestation
(D) Using natural fertilizers

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.